

Why P_b and the Starlight table disagree at large y : investigation notes

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1 Summary

The photon flux computed by this code (labelled P_b , following the Baur–Hencken–Trautmann point-like equivalent-photon formula used in our paper’s Eq. (2)) disagrees with Petja Paakkinen’s tabulated “Starlight” flux (`inputs/photon_flux/log-flux-tbl-WS.dta`, citing Eskola et al., arXiv:2404.09731) by up to two orders of magnitude at $y = 2\omega/\sqrt{s_{NN}} \gtrsim 0.05$, growing worse with y (Fig. 1). We tested three candidate explanations in turn and ruled out the first two before confirming the third.

2 Candidate 1: the electromagnetic-dissociation (EMD) modeling — ruled out

The AnAn channel carries *no* EMD factor at all (`emd_factor=1` in `photon_flux()`), while An0n applies $P(b_\perp) = \exp(-S/b_\perp^2)$. If the EMD parametrization were the source of the disagreement, AnAn should match the table closely while An0n alone diverges. Instead the two channels diverge from the table by essentially the same factor at every y (e.g. 0.030 vs. 0.030 at $y = 0.1$, Table 1) — a channel that doesn’t even use EMD tracks the one that does. EMD cannot be the explanation.

y	AnAn/table (no EMD)	An0n/table (with EMD)
10^{-4}	0.992	0.992
10^{-2}	0.902	0.902
3×10^{-2}	0.602	0.603
5×10^{-2}	0.313	0.313
10^{-1}	0.030	0.030

Table 1: Ratio of this code’s simple single- b flux to the tabulated Starlight flux, for the channel with no EMD factor (AnAn) and the one with one (An0n). They move together.

3 Candidate 2: the source nucleus’s own charge form factor — ruled out

The bare flux used throughout this code,

$$n(\omega, b_\perp) = \frac{\alpha_{em} Z^2 \omega^2}{\pi^2 \gamma^2} 2K_1(\eta)^2, \quad \eta = \frac{\omega b_\perp}{\gamma}, \quad (1)$$

is the *point-like* approximation (our paper’s Eq. (2)). The tabulated Starlight flux instead uses the Woods–Saxon form,

$$f_{\gamma/A}(z, b) = \frac{\alpha_{em} Z^2}{\pi^2 z} \left[\int_0^\infty dk_\perp \frac{k_\perp^2 F_A(\sqrt{k_\perp^2 + z^2 m_N^2})}{k_\perp^2 + z^2 m_N^2} J_1(k_\perp b) \right]^2, \quad (2)$$

with F_A the nuclear charge form factor (Eq. 16 of Guzey–Innocenti–Staśto–Strikman, arXiv:2606.05469, citing Vidović–Greiner–Best–Soff; structurally the same as Eq. 11 of Eskola et al., citing Krauss–Greiner–Soff). We implemented Eq. (2) (`flux_density_WS()`, `src/photon_flux.hpp/.cpp`) and found, both analytically (an EPA analogue of Newton’s shell theorem) and numerically (matching a brute-force independent integration to 8 significant figures), that Eq. (2) is *identical* to Eq. (1) for $b \geq R_A \approx 33 \text{ GeV}^{-1}$, and only differs for $b < R_A$ (Table 2) — the field outside a spherically symmetric charge distribution doesn’t depend on its internal structure.

$b \text{ [GeV}^{-1}]$	I_{WS}/I_{PL}	
5	0.030	(inside nucleus, $R_A = 32.9$)
20	0.446	(inside)
30	0.841	(inside)
72	1.0000	(outside; this code’s actual $b_{\min}$)
100	1.0000	(outside)

Table 2: Ratio of the Woods-Saxon to point-like $k_\perp$ -integral, at fixed $z = 0.01$, as a function of b .

Since this code’s nuclear-overlap cut is $b_{\min} = 2R_A = 14.2 \text{ fm} \approx 72 \text{ GeV}^{-1}$ — always larger than R_A — the bare-flux choice (point-like vs. Woods-Saxon) provably cannot change any cross section computed by this code. Confirmed end-to-end: `FLUX_MODEL=PL` and `FLUX_MODEL=WS` give bit-identical output from the D0 binary.

4 Candidate 3: the target-nucleus geometric convolution — confirmed

Both this code and our own paper use the *single impact-parameter* treatment,

$$N(\omega) = \int d^2b \, n(\omega, b) \Gamma_{AA}(b) P_{EMD}(b), \quad (3)$$

evaluating the bare flux and the nuclear survival probability $\Gamma_{AA}(b)$ at the *same* b . Guzey et al. (2606.05469, Eq. 15) use the identical structure, and explicitly flag it as an approximation to a fuller treatment (citing Eskola et al., ref. [16] there):

“One can in principle refine Eq. (15) by taking into account details of the transverse-plane geometry of UPC events and the spatial dependence [of] nuclear PDFs [16]. While it somewhat increases $f_{\gamma/A}(z)$ for large z , the photon flux is suppressed by several orders of magnitude in this region.”

That fuller treatment is Eskola et al.’s effective flux (Eq. 4 of 2404.09731),

$$f_{\gamma/A}^{\text{eff}}(z) = \frac{1}{B} \int d^2r \int d^2s \, f_{\gamma/A}(z, r) T_B(s) \Gamma_{AB}(r - s), \quad (4)$$

which keeps the photon’s emission point r (relative to nucleus A) and the interaction point s (spread over the target’s actual thickness T_B) as independent variables, with $b = r - s$ the true nuclear separation. Eq. (3) is the limit of Eq. (4) in which B is treated as a point, $T_B(s) = \delta^2(s)$.

We implemented Eq. (4) as a standalone diagnostic (`src/scan_flux_eff.cpp`) and productionized it into the physics pipeline (`src/photon_flux.hpp/.cpp`: `WSThickness`, `EffFluxRadial`, `init_effective_flux()`, `effective_photon_flux()`; wired into `integrand.cpp`, `int_exclusive.cpp`, `int_diffractive.cpp`, `main.cpp/main_xpom.cpp` as `FLUX_MODEL=EFF`, now the *default*). The result closes the gap dramatically (Table 3, Fig. 2): agreement improves from a factor of ~ 33 at $y = 0.1$ down to $\sim 10\%$, and stays within $O(1)$ across the whole physically relevant range, instead of collapsing. The same implementation was ported to `inclusive-D0-UPC` (`src/photon_flux.hpp/.cpp`, `inclusive_spectrum.cpp`, `main.cpp`), with one required change: that codebase’s `flux_density()` uses a (*pref*/*qp*) normalization convention rather than diffractive-D0-UPC’s (*pref*/ ω) – since $qp = \omega\sqrt{2}$, the $F_A=1$ matching condition there gives a prefactor of $\sqrt{2}/(z \cdot ss)$ instead of $2/(z \cdot ss)$. Confirmed by regression (PL/WS bit-identical to pre-change output) and by an independent physics check: at $p_{D0\perp} = 10$ GeV, $y = 2$, $\text{EFF/PL} = 1.40$ for the inclusive spectrum, consistent with (and slightly larger than, as expected with no competing exclusive/diffractive split) the ~ 1.3 found in the diffractive channels at comparable kinematics.

4.1 An exact closed-form nuclear form factor

The appendix of Eskola et al. (arXiv:2404.09731) explains their own numerical implementation of Eq. (2):

“the integrand in Eq. (11) oscillates rapidly at large $|r|$ and the integral is therefore poorly converging. For this, we use the result that at large distances the bare flux always reduces to that of a PL source and substitute for the large- $|r|$ bare flux $f_{\gamma/A}^{WS}(y, |r| > 2R_{PL}) = f_{\gamma/A}^{PL}(y, r)$. Furthermore, for smaller $|r|$ we reduce the number of numerical integrations by using the analytic expression for the Fourier [transform]...”

This is the same large- r PL-fallback trick used in `flux_density_WS()`, though at a different switch point ($2R_{PL} \approx 72 \text{ GeV}^{-1}$ vs. this code’s $R_A \approx 33 \text{ GeV}^{-1}$; both are numerically safe, verified against WS/PL convergence directly). More importantly, for small $|r|$ they use an *analytic* closed form for $F_A(q)$ rather than a brute-force numerical Fourier transform (which is what `WSFormFactor` originally did here). That analytic form is due to Maximon & Schrack, “*The form factor of the Fermi model spatial distribution*,” J. Res. Natl. Bur. Stand. B **70**, 85 (1966), their Eq. (20): for $\rho_A(r) = \rho_0/(1 + e^{(r-c)/a})$,

$$F_A(q) = \frac{4\pi^2\rho_0a^3}{(qa)^2\sinh^2(\pi qa)} \left[\pi qa \cosh(\pi qa) \sin(qc) - qc \cos(qc) \sinh(\pi qa) \right] + 8\pi\rho_0a^3 \sum_{n=1}^{\infty} (-1)^{n-1} \frac{n e^{-nc/a}}{[n^2 + (qa)^2]^2}, \quad (5)$$

with ρ_0 fixed by their Eq. (21) so that $F_A(0) = 1$ exactly. We replaced `WSFormFactor`’s internals with Eq. (5): for lead, $c/a \approx 12$, so the series converges to double precision in a handful of terms, needing no grid or spline at all. The new values agree with the old brute-force numerical ones to 5–6 significant figures (the old implementation was already accurate), so this did not measurably move the residual wiggle described below – but it is now exact rather than approximate, removes a class of possible numerical artifacts, and matches the method the reference table itself was likely built with.

y	simple/table (Eq. 3)	effective/table (Eq. 4)	AnAn channel
10^{-4}	0.99	1.00	
10^{-2}	0.90	1.00	
3×10^{-2}	0.60	1.01	
5×10^{-2}	0.31	1.02	
10^{-1}	0.030	1.10	

Table 3: Ratio to the tabulated Starlight flux, simple vs. effective treatment. An0n channel shows the same pattern to within 1–2%.

4.2 Impact on physical cross sections

Since b is integrated out entirely under Eq. (4), the fraction of the cross section coming from $y_\gamma > 0.01$ (where the simple and effective treatments diverge) sets the size of the correction. That fraction is negligible at low $p_{D0\perp}$ /backward rapidity but reaches ~ 60 –100% at high $p_{D0\perp}$ /forward rapidity. Correspondingly, switching from `FLUX_MODEL=PL` to `FLUX_MODEL=EFF` changes both the exclusive and diffractive D^0 cross sections by:

$p_{D0\perp}$ [GeV]	y	exclusive (EFF/PL)	diffractive (EFF/PL)
2	−2	1.05	1.01
3	0	1.03	1.01
8	2	1.31	1.30

i.e. a genuine, kinematics-dependent effect of up to $\sim 30\%$ at the high- $p_\perp$ /forward-rapidity end of the phase space explored in this analysis, not a correction that can be neglected there.

5 A residual feature: diffractive-like minima at large y

Even under the effective-flux treatment, a residual $O(1)$ wiggle in the ratio to the table remains around $y \sim 0.15$ –0.3 (visible in Fig. 2). We traced roughly half an order of magnitude of this to a numerical artifact — the asymptotic matching radius used to cap the geometric convolution’s numerical grid (R_{switch}) was too small, giving a $\sim 0.7\%$ discontinuity there for the EMD-bearing channels; pushing R_{switch} from 300 to 800 GeV^{-1} (and switching to a log-spaced radial grid) cut that discontinuity to $\sim 0.1\%$ without changing the wiggle’s amplitude. The remaining feature is consistent with a phenomenon both source papers describe explicitly. Eskola et al. (2404.09731), discussing their own bare-flux comparison (PL vs. Woods-Saxon, their Fig. 2(a)), write:

“As is well known, the differences to the PL approximation reside almost exclusively in the region within the emitting nucleus. Only for the very high energy photons where additional diffractive-like minima appear due to the integration over the form factor and the three approximations begin to divert.”

i.e. the nuclear form factor F_A entering the $k_\perp$ integral (Eq. (2)) develops its own diffraction pattern (a zero crossing around $q \sim 0.1$ –0.2 GeV , confirmed directly from our own `WSFormFactor` implementation) at high photon energy, and small differences in exactly how that integral is discretized between two independently-written codes can shift the location of such a sharp feature enough to produce an $O(1)$ ratio bump even though both underlying curves are basically correct. Both source papers dismiss this regime as practically unimportant, since the flux itself is suppressed by several orders of magnitude there.

6 Files

- `src/photon_flux.hpp`, `src/photon_flux.cpp` — `WSFormFactor` (exact, Eq. 5), `WSFluxTable` (Eq. 2); `WSThickness`, `EffFluxRadial`, `init_effective_flux()`, `effective_photon_flux()` (Eq. 4).
- `src/scan_flux_eff.cpp` — standalone diagnostic that validated Eq. (4) before it was productionized.
- `src/integrand.cpp`, `src/int_exclusive.cpp`, `src/int_diffraction.cpp`, `src/main.cpp`, `src/main_xpom.cpp` — `FLUX_MODEL` now defaults to `EFF`; `PL`/`WS` remain available for comparison/reproducing old results.
- `python/flux_comparison.py`, `plots/flux_comparison.pdf` — simple-treatment comparison to the Starlight table (Fig. 1).
- `python/eff_flux_check.py`, `plots/eff_flux_check.pdf` — simple vs. effective comparison (Fig. 2).
- `../inclusive-D0-UPC/src/photon_flux.hpp/.cpp`, `inclusive_spectrum.cpp`, `main.cpp` — the same implementation ported to the inclusive-spectrum codebase (see §4 for the normalization-convention difference this required).

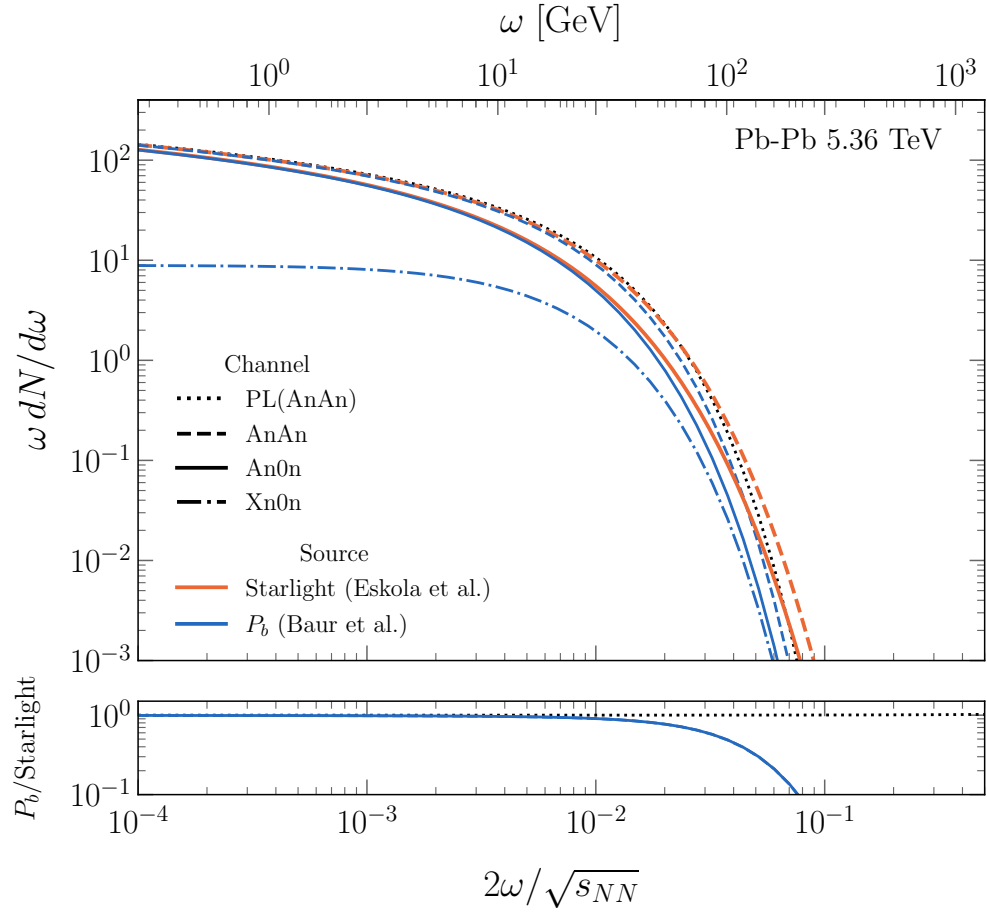


Figure 1: This code's simple single- b flux (Eq. 3) vs. the tabulated Starlight flux.

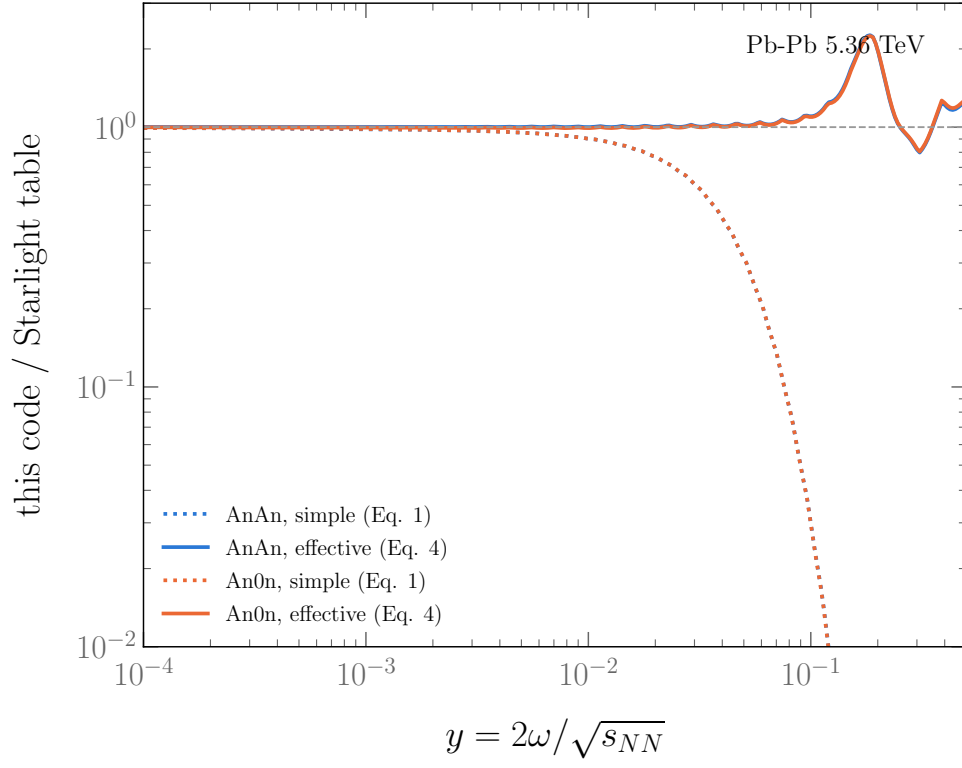


Figure 2: Ratio to the Starlight table: simple (dotted) vs. effective-flux (solid) treatment, for the AnAn and An0n channels.